

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.23	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.9	Product Info 2	A1
Occupation	WORKER	Annual Income	\$305,000.00
Insurance Type	Life	Coverage Amount	\$4,500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Life Insurance Application Assessment

Date: October 26, 2023

Applicant ID: [Not Provided, referring to Current Applicant]

Insurance Type: Life

Coverage Amount: \$4,500,000.00

1. Risk Assessment

The applicant presents a complex risk profile with significant inconsistencies:

* **Age:** 0.23 years (approximately 2.7 months old). This is the most critical and concerning factor.

* **BMI:** 22.9 kg/m², derived from height 175.0 cm and weight 70.0 kg. This BMI is within a healthy range for an adult, but entirely inconsistent with a 0.23-year-old infant.

* **Smoking Status:** Smoker (1). This is a major risk factor and is fundamentally incompatible with the stated age of 0.23 years. An infant cannot be a smoker.

* **Occupation:** 'WORKER'. This occupation is incompatible with an infant.

* **Income:** \$305,000.00. This high income is incompatible with an infant.

* **Previous Claims:** 0. Favorable.

* **Family History:** 0. Favorable.

* **Lifestyle:** Exercise (0), Alcohol (0). Assuming these mean 'no' or 'rarely', they are generally favorable for an adult, but again, irrelevant for an infant.

* **Coverage Amount:** \$4,500,000.00. This is a very substantial coverage amount. While high-net-worth parents may insure children, this level combined with the other inconsistencies raises significant concerns.

Summary of Risk: The stated applicant details create a highly anomalous profile, where the age (infant) contradicts fundamental risk factors like smoking, occupation, income, and adult-like physical characteristics (height, weight, BMI).

2. Comparative Analysis

The machine learning model predicted a Risk Class of 1, with a calculated annual premium of \$10,000.00.

The top similar historical approved cases are all flagged with 100.0% similarity, yet present higher Risk Classes and Premiums:

* **POL-14102:** Risk Class 2, Premium \$12,000.00

* **POL-21636:** Risk Class 2, Premium \$12,000.00

* **POL-52224:** Risk Class 3, Premium \$15,000.00

Comparison Points:

* **Similarity Match:** The current application is deemed 100% similar to these historical cases. This implies that, based on the features used for similarity matching (likely including `product\_info\_2`, `occupation`,

`income`, `smoker`, `previous\_claims`, `family\_history`, `exercise`, `alcohol`, `gender`, and possibly `height`/`weight`/`bmi`), the profiles are identical.

* **Risk Class Discrepancy:** Despite 100% similarity in input features, the current application's predicted Risk Class (1) is significantly lower (more favorable) than the historical cases (2 and 3). This divergence suggests one of the following:

* The underlying machine learning model for risk classification has been updated and is now more optimistic or accurate.

* The "similarity" metric for RAG uses a subset of features, and a key distinguishing factor not captured by the RAG similarity led to the lower predicted risk class for the current applicant.

* There is an inconsistency between the RAG system's similarity calculation and the ML classifier's output for what appear to be identical profiles.

* **Premium Discrepancy:** The lower calculated premium (\$10,000.00) for the current applicant logically follows from its more favorable predicted Risk Class (1) compared to the higher premiums for Risk Class 2 and 3 in historical cases.

3. Anomaly & Fraud Detection

Several critical anomalies are present, suggesting potential data entry error or misrepresentation:

* **Age-Smoking Inconsistency:** The most glaring anomaly is an applicant aged 0.23 years (an infant) being identified as a smoker. This is biologically impossible and indicates a fundamental data error.

* **Age-Income/Occupation Inconsistency:** An infant cannot have an income of \$305,000.00 or be classified with the occupation 'WORKER'.

* **Age-Physical Attributes Inconsistency:** The height (175 cm) and weight (70 kg) resulting in a BMI of 22.9 are adult measurements, completely inconsistent with an infant.

* **High Coverage for Inconsistent Profile:** A \$4.5 million coverage amount for an applicant with such contradictory details is highly suspicious.

* **ML Prediction Confidence:** The prediction confidence of 60.1% is moderate, which is not exceptionally high, especially given the contradictory input data.

* **RAG Similarity vs. Risk Class Discrepancy:** While not necessarily an anomaly in itself, the 100% similarity to higher-risk historical cases, yet a lower predicted risk class, warrants further investigation if the core data inconsistencies were resolved. It could indicate a model drift or feature weighting difference.

These anomalies strongly suggest that the applicant's age has been entered incorrectly, or that details from different individuals (e.g., an adult policyholder and an infant beneficiary) have been erroneously combined. This could represent a significant underwriting risk due to misrepresented information.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 is aligned with the predicted Risk Class 1, which is the lowest risk category. Historically, similar (but higher risk class) approved applications commanded higher premiums (\$12,000 - \$15,000). If the predicted Risk Class 1 were accurate and based on valid data, the \$10,000.00 premium would be logically justified given the relative risk.

However, given the severe inconsistencies in the applicant's profile (age vs. smoking, occupation, income, and physical attributes), **the current premium cannot be justified or accepted.** The input data used for the ML classification is highly likely to be erroneous. A premium based on these flawed inputs would not reflect the true risk.

5. Underwriting Decision Recommendation

Recommendation: Refer for Manual Review and Data Verification

Due to the critical and multiple inconsistencies in the applicant's profile, a direct approve or reject decision is inappropriate. The application must be immediately flagged and referred for a comprehensive manual review.

Specific actions required:

1. **Verify Applicant Identity and Age:** Clarify the true age of the applicant.
2. **Verify Smoking Status:** Confirm if the true applicant is a smoker.
3. **Verify Occupation and Income:** Confirm the applicant's actual occupation and income.
4. **Recalculate BMI/Physical Attributes:** Obtain accurate height and weight for the true applicant.
5. **Address Discrepancies:** Investigate how these highly contradictory data points were entered into the system. It is highly probable that the age (0.23) is a data entry error, and the other details (smoking, occupation, income, BMI) pertain to an adult (e.g., a parent or guardian) who is the true intended policyholder, or there's an error in linking data.

Until these fundamental data discrepancies are resolved, the risk assessment, comparative analysis, and premium calculation are based on flawed premises and cannot be relied upon for an underwriting decision. Attempting to underwrite this policy based on the current data would expose the insurer to significant unquantified risk.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.